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Editor-in-Chief
Computers in Biology and Medicine
Editorial Office

Dear Editor,

We are pleased to submit our original research article entitled “**When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study**” for your consideration as an **Original Research Article** in *Computers in Biology and Medicine*. This manuscript has not been published elsewhere and is not under consideration by any other journal.

Summary of contributions. To our knowledge, this is the first study to systematically characterize both the payoff and the boundary conditions of Temperature Scaling (TS) as a post-hoc recalibration method under cross-corpus ECG transfer. We pre-registered and executed a multi-seed calibration boundary study comprising **60 experiments** (6 directed transfer pairs $\times$ 2 architectures $\times$ 5 seeds, fully balanced) on PTB-XL, Chapman-Shaoxing, and CPSC2018+2019, with patient-level cluster paired bootstrap ($B = 10,000$; percentile operating CI with BCa sensitivity on 3 representative experiments) and eight recalibration methods. TS yields a positive out-of-distribution (OOD) calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$ in **51/60 = 85.0%** of seed experiments (InceptionTime 27/30, ResNet1D 24/30), while the remaining **9 counter-examples are transparently disclosed** and mechanistically explained case by case (six of nine co-occur with low OOD accuracy; seven of nine with a large ID–OOD accuracy gap). The paper delivers **four independent contributions**: **C1** (OOD benefit quantification, 51/60 seed support), **C2** (ID boundary condition with the pre-registered branch triggered: only 29/60 = 48.3% of cells have CIs containing zero, below the $\geq 80\%$ criterion, so the “ID nonzero boundary” branch applies and C1 is reported with this qualifier), **C3** (27-cell Shapley decomposition validation, $n = 20,000$), and **C4** (method-selection boundary: TS robust vs. EM prior and Matrix scaling weakest, governed by a prior-likelihood mismatch mechanism).

Why this journal. Cross-corpus ECG transfer calibration is a critical bottleneck for clinical deployment of deep ECG models across institutions with different acquisition protocols, lead sets, and label vocabularies. Our methodological contribution—an explicit boundary-condition characterization of when TS pays off, combined with transparent reporting of nine counter-examples and a pre-registered failure branch—aligns directly with the journal’s scope on biomedical signal processing, clinical machine learning reliability, and reproducible computational methods in medicine.

Transparency statement. All hypotheses in this manuscript are tagged *exploratory + robustness validation*, not confirmatory, consistent with the open-science best practices of Nosek et al. (2018). The nine counter-examples (15.0% of the experiment grid) are disclosed in full in the main text and

supplements; no non-supporting result is omitted or post-hoc excluded. The pre-registration protocol v2.1-A1, its hash manifest, and revision A1 are included with the submission as supplementary files; public OSF archival is in progress and the hash manifest allows independent verification of the snapshot timing. All deviations from the original protocol are recorded in the formal revision A1. The source code, experiment artifacts, and data preparation scripts are provided with the submission as supplementary material; a public repository will accompany acceptance. A five-item honesty declaration (adversarial audit product) is included in the manuscript and archived with the protocol.

Note on pre-registration design. This study adopts an *exploratory + robustness validation* design rather than a strictly confirmatory design. The pre-registration protocol was made public before the multi-seed validation was unblinded. The formal revision A1 records a transparent re-location of the primary endpoint from the original “ID→OOD decay” to “OOD benefit + boundary conditions”. This re-location was triggered not by the main-test results but by an independent exploratory pilot ($\Delta\text{ECE}_{\text{ID}} \approx 0$), and it is consistent with the empirically observed ID–OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019); we emphasize this asymmetry is empirical, not a theorem. Following Lakens (2021) and Nosek et al. (2018), this constitutes a **transparent re-location, not HARKing**: the pilot results do not enter the main test report, and the revision direction is consistent with the registered empirical asymmetry rather than result-driven. We disclose this design choice at the outset so that reviewers and editors can evaluate the evidence under the correct inferential framework.

We believe this work makes a significant methodological contribution to the field of cross-corpus ECG calibration, providing an actionable answer to “when does TS pay off” with multi-seed robustness evidence, transparent counter-example reporting, and a pre-registered boundary-condition analysis. We look forward to your feedback.

Sincerely,

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